

Section A:

1) $x \leq 3$ and $x \leq 2 \Rightarrow x \leq 2$

2) $\lim_{x \rightarrow \infty} 2 \tan^{-1} 2x = 2 \lim_{x \rightarrow \infty} \tan^{-1}(2x)$
 $= 2 \lim_{x \rightarrow \infty} \tan^{-1}(x) = 2 \times \frac{\pi}{2} = \pi.$

3) Statement (1): Correct

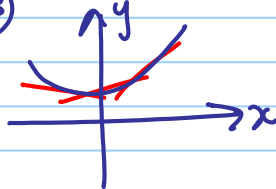
Statement (2): False, since $0 \notin D_f$

Statement (3): Correct, since $0 \in D_f$.

4) $\lim_{x \rightarrow -5^+} \frac{x-3}{x+5} \left(\frac{-8}{0^+} \right)$
 $= -\infty$

$x \rightarrow -5^+$
 $\Rightarrow x > -5$
 $\Rightarrow x+5 > 0$

- 5) (1) see p. 302 of textbook
 (2) see p. 302 of textbook
 (3)



graph above tangent line
and graph concave up

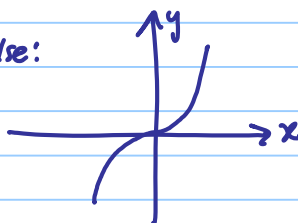
- 6) (1) False: Counterexample: $f(x) = x+3$
 $g(x) = x+4$

(2) False:



Local max at $x=c$
 but $f'(c)$ does not exist

(3) False:



$f(x) = x^3$
 $f'(0) = 0$ but no local extreme at $x=0$.

7)

7) (1) False

(2) True. $f(x) = \sinh 3x + \cosh 3x$
 $f'(x) = 3 \cosh 3x + 3 \sinh 3x$
 $= 3(\sinh 3x + \cosh 3x)$

(3) Correct $f(x) = e^{3x}$
 $f'(x) = 3e^{3x}$
 $f'(-x) = 3e^{-3x}$

(4) $\frac{\infty}{0}$ is not an indeterminate form

$$\lim_{x \rightarrow 0} \frac{x}{e^x} = \frac{0}{e^0} = \frac{0}{1} = 0.$$

$$\lim_{x \rightarrow -\infty} \frac{x}{e^x} = \lim_{x \rightarrow -\infty} x e^{-x} \quad (-\infty \times \infty)$$
$$= -\infty$$

Correct options

- (1) b
- (2) c
- (3) e
- (4) d

- (5) g
- (6) h
- (7) f
- (8) h.

Memo

SECTION B [Fill in questions] - 13 marks

SECTIONS B AND C MUST BE SCANNED AND UPLOADED ON CLICKUP.

Please do not scan your rough working or any empty pages.

SURNAME:
INITIALS:
STUDENT NUMBER:

SECTION B: Write only your final, fully simplified answer in the block provided. Only this will be marked. Rough working can be done outside the blocks and will not be marked.

QUESTION 9 [1 mark]

If $f(1) = 5$ and $f'(x) \geq 2$ for $1 \leq x \leq 5$, write down the smallest possible value of $f(4)$.

11

$$f(4) = 5 + 2(3) = 5 + 6 = 11$$

QUESTION 10 [1 mark]

Suppose $f(2) = 5$, $g(2) = 6$, $f'(2) = 1$ and $g'(2) = -3$. Then $(fg)'(2) =$

-9

$$(fg)'(2) = f'(2)g(2) + f(2)g'(2) = 1(6) + 5(-3) = 6 - 15 = -9$$

QUESTION 11 [1 mark]

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} = \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{(\sqrt{x} - 3)(\sqrt{x} + 3)} = \frac{1}{3 + 3}$$

1/6

QUESTION 12 [1 mark]

$$\frac{d}{dt}(\ln t)^2 =$$

$$= 2(\ln t) \cdot \frac{1}{t}$$

$\frac{2 \ln t}{t}$

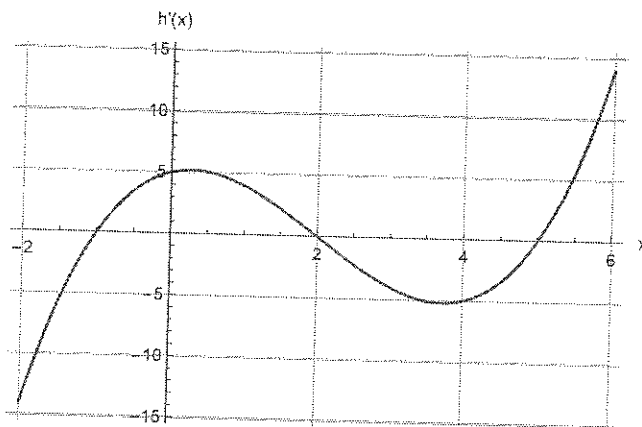
QUESTION 13 [1 mark]

The vertical asymptote(s) of the function $f(x) = \frac{x^2 - 5x}{x^3 - 5x^2 + 6x}$ is(are):

$x = 3$ and $x = 2$.

$$f(x) = \frac{x(x-5)}{x(x^2-5x+6)} = \frac{x-5}{(x-3)(x-2)}$$

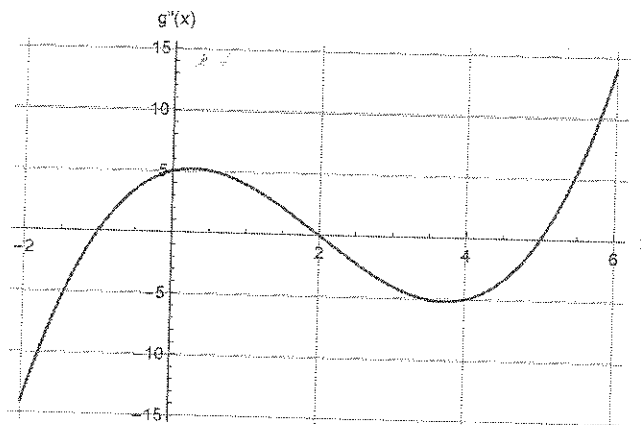
QUESTION 14 [2 marks]

(a) Consider the graph of h' :Write down the interval(s) on which the function h is increasing.

(1)

$$(-1, 2) \cup (5, \infty)$$

$$(-1, 2) \text{ and } (5, \infty)$$

(b) Consider the graph of g'' :Write down the interval(s) on which the function g is concave down.

(1)

$$(-\infty, -1) \text{ and } (2, 5)$$

$$\frac{dy}{dx} = \frac{(x^2+1)(-e^{-x}) - e^{-x}(2x)}{(x^2+1)^2}$$

QUESTION 15 [2 marks]

The point on the curve $y = \frac{e^{-x}}{x^2+1}$ at which the tangent line is horizontal, is

$$(-1, \frac{e}{2})$$

QUESTION 16 [2 marks]

$$\lim_{\theta \rightarrow 0} \frac{\tan(4\theta)}{4\theta + \sin \theta} = \left(\frac{0}{0}\right)$$

$$= \lim_{\theta \rightarrow 0} \frac{4 \sec^2 4\theta}{4 + \cos \theta}$$

$$= \frac{4}{4+1}$$

$$4/5$$

$$= \frac{-e^{-x}(x^2+2x+1)}{(x^2+1)^2}$$

$$= \frac{-e^{-x}(x+1)^2}{(x^2+1)^2}$$

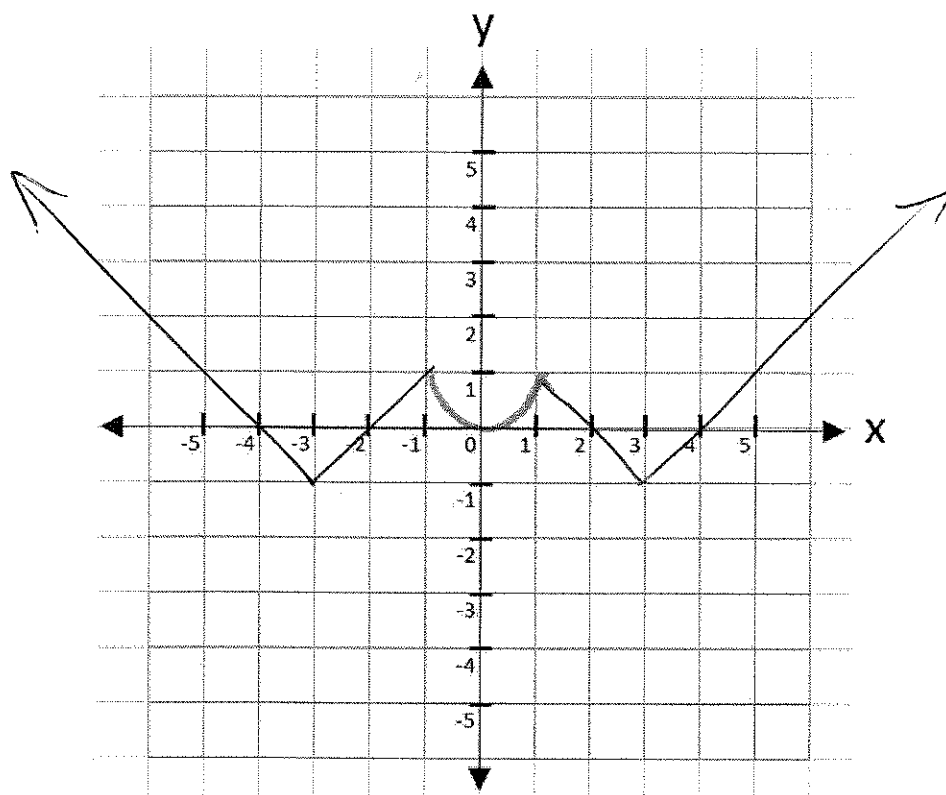
$$= 0 \Rightarrow x = -1$$

$$y = \frac{e}{1+1} = \frac{e}{2}$$

QUESTION 17 [2 marks]

Sketch the graph of the function f on the axes provided, indicating all relevant intercepts and information, where f is continuous and even and

$$f(0) = 0, f'(x) = 2x \text{ if } 0 < x < 1, f'(x) = -1 \text{ if } 1 < x < 3 \text{ and } f'(x) = 1 \text{ if } x > 3.$$



SECTION C [Full written answers] - 20 marks

QUESTION 18 [4 marks]

Consider the function $f(x) = \begin{cases} ax + b & \text{if } x < 1 \\ 3 & \text{if } x = 1 \\ bx^2 - a & \text{if } x > 1 \end{cases}$

- (a) Find the values of a and b for which $\lim_{x \rightarrow 1} f(x)$ exists. Explain your answer.

(2)

$$\lim_{x \rightarrow 1} f(x) \text{ exists if } \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (ax + b) = a + b$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (bx^2 - a) = b - a$$

$$\Rightarrow \lim_{x \rightarrow 1} f(x) \text{ exists if } a + b = b - a$$

$$\Rightarrow a = -a$$

$$\Rightarrow 2a = 0 \Rightarrow a = 0$$

$$\therefore \underline{a = 0 \text{ and } b \in \mathbb{R}}$$

- (b) Complete the definition:

The function f is continuous at $x = 1$ if

(1)

$$\lim_{x \rightarrow 1} f(x) = f(1)$$

- (c) Find the values of a and b for which the function $y = f(x)$ is continuous at $x = 1$.

(1)

$$\lim_{x \rightarrow 1} f(x) \text{ exists if } a = 0 \text{ and } b \in \mathbb{R}$$

$$\underline{f \text{ continuous}} \text{ if } \lim_{x \rightarrow 1} f(x) = f(1)$$

$$\Rightarrow a + b = 3,$$

$$\text{But } a = 0$$

$$\Rightarrow b = 3$$

$$\therefore a = 0 \text{ and } b = 3.$$

QUESTION 19 [5 marks]

Determine the limit, if it exist(s).

(a) $\lim_{x \rightarrow -\infty} \sqrt{x^2 - 3x} + x$ $(\infty - \infty)$ (3)

$$= \lim_{x \rightarrow -\infty} \frac{(\sqrt{x^2 - 3x} + x)(\sqrt{x^2 - 3x} - x)}{\sqrt{x^2 - 3x} - x}$$

$$= \lim_{x \rightarrow -\infty} \frac{x^2 - 3x - x^2}{\sqrt{x^2 - 3x} - x}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3x}{\sqrt{x^2 - 3x} - x}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3x}{|x^2|^{1/2} \sqrt{1 - 3/x} - x}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3x}{-x \sqrt{1 - 3/x} - x}$$

$$= \lim_{x \rightarrow -\infty} \frac{3}{\sqrt{1 - 3/x} + 1}$$

$$= \frac{3}{2}$$

(b) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{2\sqrt{x}}$ $(\frac{\infty}{\infty})$ Form NB

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{2(\frac{1}{2} x^{-1/2})}$$

(L'Hospital)

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{\sqrt{x}}}$$

Simplify!

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0$$

NB

$$\sqrt{x^2} = |x| = -x \text{ if } x < 0$$

(2)

NB to mention.

C

QUESTION 20 [3 marks]

Use the Closed Interval Method to determine the extreme values of $f(x) = \frac{1}{x^2 + x + 1}$ on the interval $[-2, 1]$.

$$f(x) = (x^2 + x + 1)^{-1}$$

$$f'(x) = -1(x^2 + x + 1)^{-2} \times (2x + 1)$$

$$= \frac{-(2x + 1)}{(x^2 + x + 1)^2}$$

$$f'(x) = 0 \Rightarrow 2x + 1 = 0$$

$$\Rightarrow x = -\frac{1}{2}$$

$$f(-2) = \frac{1}{4 - 2 + 1} = \frac{1}{3}$$

$$f\left(-\frac{1}{2}\right) = \frac{1}{\frac{1}{4} - \frac{1}{2} + 1} = \frac{4}{3}$$

$$f(1) = \frac{1}{1 + 1 + 1} = \frac{1}{3}$$

Local minimum: $\frac{1}{3}$ at $x = -2$ and $x = 1$

Local maximum: $\frac{4}{3}$ at $x = -\frac{1}{2}$.

The function value.

The function value.

QUESTION 21 [3 marks]

We know that $\sin^{-1}(x) = y$ if and only if $\sin y = x$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

- (a) Use implicit differentiation and the definition of the function $\sin^{-1}(x)$ to show that

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}. \quad (2)$$

See page. 222.

- (b) Find $\frac{dy}{dx}$ if $y = \sin^{-1}(\tan x)$.

(1)

See p 222

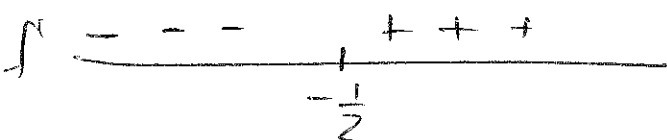
Also apply chain rule, as in
Example 9 b.

QUESTION 22 [3 marks]

Use the First Derivative Test to determine the local extreme values of the function $f(x) = xe^{2x}$.

$$f'(x) = e^{2x} + 2xe^{2x} \\ = (2x+1)e^{2x}$$

$$f'(x) = 0 \Rightarrow 2x+1=0 \\ \Rightarrow x = -\frac{1}{2}$$



$$f'(-10) < 0$$

$$f'(0) > 0$$

$\Rightarrow f$ has a local minimum of $f(-\frac{1}{2}) = -\frac{1}{2}e^{-1} = -\frac{e}{2}$
at $x = -\frac{1}{2}$.

QUESTION 23 [2 marks]

Let f and g be differentiable functions with $f > 0$ and suppose $g = \frac{1}{f}$. Show that g is concave up when f is concave down.

Given: $f''(x) < 0$ and $f > 0$

RTP: g is concave up, $\therefore g''(x) > 0$

$$g'(x) = \frac{d}{dx} (f(x))^{-1} = -1 f(x)^{-2} \times f'(x) = \frac{-f'(x)}{f(x)^2}$$

$$g''(x) = \frac{-f''(x)f(x)^2 - 2f(x)f'(x) \times (-f'(x))}{f(x)^4} \\ = \frac{-f''(x)f(x)^2 + 2f(x)f'(x)^2}{f(x)^4}$$

Since $f''(x) < 0$ and $f(x) > 0$, we know that

$$g''(x) > 0$$

End of Section C

$\Rightarrow g$ is concave up.